

Math 526, S05
Quiz 2

Name:
SSN: xxx-xx-

Instructions: There are totally 1 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 0 & 1 & -1 \\ 1 & -1 & 1 \\ -2 & 4 & -2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -1 \\ 0 \\ 10 \end{bmatrix}$.

a) Find the $\mathbf{L} - \mathbf{U} - \mathbf{P}$ factorization of $\mathbf{A}$.

Solution:

$$\begin{array}{ccc} \left[\begin{array}{c} \\ \\ \end{array} \right] & \left[\begin{array}{ccc} 0 & 1 & -1 \\ 1 & -1 & 1 \\ -2 & 4 & -2 \end{array} \right] & \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \\ \left[\begin{array}{c} \\ \\ \end{array} \right] & \left[\begin{array}{ccc} 1 & -1 & 1 \\ 0 & 1 & -1 \\ -2 & 4 & -2 \end{array} \right] & \left[\begin{array}{ccc} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array} \right] \\ \left[\begin{array}{c} 0 \\ -2 \\ \end{array} \right] & \left[\begin{array}{ccc} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 2 & 0 \end{array} \right] & \left[\begin{array}{ccc} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array} \right] \\ \left[\begin{array}{cc} 0 & 2 \\ -2 & 2 \end{array} \right] & \left[\begin{array}{ccc} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{array} \right] & \left[\begin{array}{ccc} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array} \right] \end{array}$$

So we have

$$\mathbf{L} = \begin{bmatrix} 1 & & \\ 0 & 1 & \\ -2 & 2 & 1 \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

b) Using the $\mathbf{L} - \mathbf{U} - \mathbf{P}$ factorization from (a) to solve $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

Solution:

First step: Solve $\mathbf{Ly} = \mathbf{Pb}$. It is easy to find

$$\mathbf{Pb} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 10 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 10 \end{bmatrix}$$

So we solve

$$\begin{bmatrix} 1 & & \\ & 1 & \\ -2 & 2 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 10 \end{bmatrix}$$

We get

$$y_1 = 0, \quad y_2 = -1, \quad y_3 = 12.$$

Second step: Solve $\mathbf{Ux} = \mathbf{y}$.

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 12 \end{bmatrix}$$

We have

$$x_3 = 6, \quad x_2 = 5, \quad x_1 = -1.$$

So we have

$$\mathbf{x} = \begin{bmatrix} -1 \\ 5 \\ 6 \end{bmatrix}.$$